

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Data Science	Course Code:	CS4048
Program:	BS (CS) and BS (SE)	Semester:	Spring 2026
Duration:	60 Minutes	Total Marks:	100
Paper Date:	April 8, 2026	Weight	15 %
Section:	All Sections	Page(s):	4
Exam Type:	Mid Term II		

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BSE-80A

Instruction/Notes: Write concise answers. Don't fill the table titled Questions/Marks.

Question	1	2	3	Total
Marks	/40	/30	/30	/100

Q1. a. What are the methods to find the value of k in k-means/k-medoids algorithm? Draw and explain the importance of Dendrogram tree? (10 + 6 marks)

b. Show the steps of **FP-Growth Algorithm** on the following data and generate rules. There are 9 items A1 to A9 (9 columns) and 15 transactions (rows). The support count is 3. (24 marks)

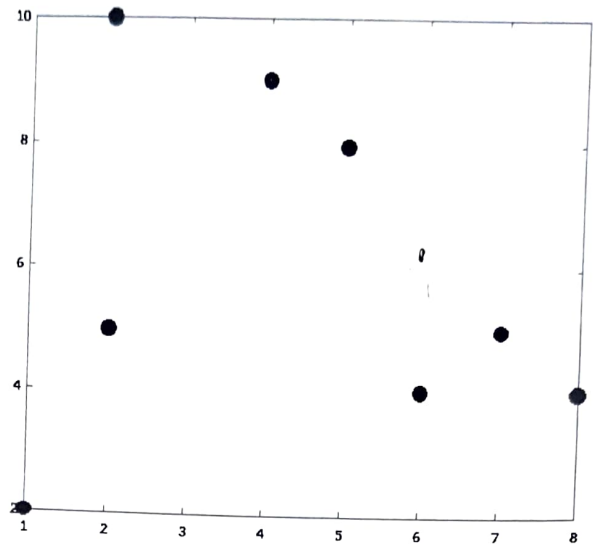
	A1	A2	A3	A4	A5	A6	A7	A8	A9
1	1	0	0	0	1	1	0	1	0
2	0	1	0	1	0	0	0	1	0
3	0	0	0	1	1	0	1	0	0
4	0	1	1	0	0	0	0	0	0
5	0	0	0	0	1	1	1	0	0
6	0	1	1	1	0	0	0	0	0
7	0	1	0	0	0	1	1	0	1
8	0	0	0	0	1	0	0	0	0
9	0	0	0	0	0	0	0	1	0
10	0	0	1	0	1	0	1	0	0
11	0	0	1	0	1	0	1	0	0
12	0	0	0	0	1	1	0	1	0
13	0	1	0	1	0	1	1	0	0
14	1	0	1	0	1	0	1	0	0
15	0	1	1	0	0	0	0	0	1

$A5 > A7 > A2 > A3 > A6 > A4 > A8$

Q2. For the following points use one iteration of k-medoids clustering to partition the given points in 2 clusters. (30 marks)

Points: A1 = (2, 10), A2 = (2, 5), A3 = (8, 4), A4 = (5, 8), A5 = (7, 5), A6 = (6, 4), A7 = (1, 2), A8 = (4, 9)

For your convenience, the plot of the points is shown below



Assume that the following points are the Initial centers: C1= (2, 10) and C2= (5, 8)

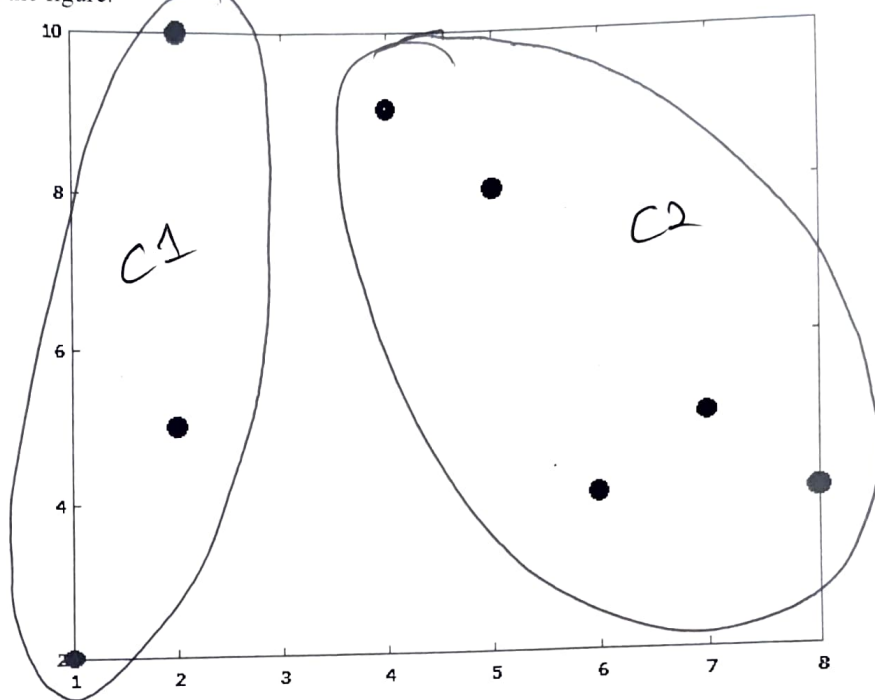
i) You have to fill the following tables with the correct values.

	Distance From C1	Distance From C2	Cluster Assignment
A1	0	5	C1
A2	5	6	C1
A3	12	7	C2
A4	5	0	C2
A5	10	5	C2
A6	10	5	C2
A7	9	10	C1
A8	3	2	C2

- ii) Using the distances computed in part i, determine the clusters

Points in first Cluster	Points in Second Cluster
A1, A2, A7 Total = 3	A3, A4, A5, A6, A8, Total = 5

- iii) Finally plot the centers on the figure below and identify the final clusters after the first iteration. [You might not compute all the distances again, just estimate the cluster from the visual distances from the figure.]



Q3. Calculate new weights and bias values by using Backpropagation Algorithm. Multi-layer feed-forward neural network is given below. The input and target values for this problem are $x_1 = 1$, $x_2 = 4$, $x_3 = 5$ and t_1 at $o_1 = 0.1$, t_2 at $o_2 = 0.05$. Initialization of weights is already given below. Learning rate of $\alpha = 0.01$. Calculate the updated weights after one iteration using Back Propagation Algorithm using a sigmoid function. (30 marks)

$$\eta = 0.01$$

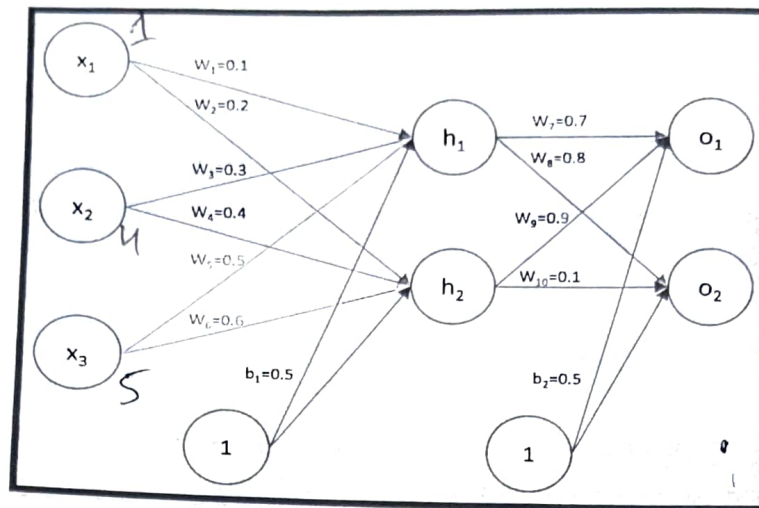


Figure 1: ANN Architecture

Sigmoid Activation Function:

$$\frac{1}{1 + e^{-x}}$$

Algorithm: Backpropagation. Neural network learning for classification or numeric prediction, using the backpropagation algorithm.

Input:

- D , a data set consisting of the training tuples and their associated target values;
- η , the learning rate;
- $network$, a multilayer feed-forward network.

Output: A trained neural network.

Method:

```

(1) Initialize all weights and biases in  $network$ ;
(2) while terminating condition is not satisfied {
(3)   for each training tuple  $X$  in  $D$  {
(4)     // Propagate the inputs forward:
(5)     for each input layer unit  $j$  {
(6)        $O_j = I_j$ ; // output of an input unit is its actual input value
(7)     }
(8)     for each hidden or output layer unit  $j$  {
(9)        $I_j = \sum_i w_{ij} O_i + \theta_j$ ; // compute the net input of unit  $j$  with respect to
        the previous layer,  $i$ 
(10)       $O_j = \frac{1}{1 + e^{-I_j}}$ ; // compute the output of each unit  $j$ 
(11)    } // Backpropagate the errors:
(12)    for each unit  $j$  in the output layer
(13)       $Err_j = O_j(1 - O_j)(T_j - O_j)$ ; // compute the error
(14)    for each unit  $j$  in the hidden layers, from the last to the first hidden layer
(15)       $Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk}$ ; // compute the error with respect to
        the next higher layer,  $k$ 
(16)    for each weight  $w_{ij}$  in  $network$  {
(17)       $\Delta w_{ij} = (\eta) Err_j O_i$ ; // weight increment
(18)       $w_{ij} = w_{ij} + \Delta w_{ij}$ ; // weight update
(19)    }
(20)    for each bias  $\theta_j$  in  $network$  {
(21)       $\Delta \theta_j = (\eta) Err_j$ ; // bias increment
         $\theta_j = \theta_j + \Delta \theta_j$ ; // bias update
    }
  }

```

Good Luck!